Data Cleaning & Visualization Project

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Introduction

This report is about a Data Cleaning and Visualization project where a dataset is processed to get useful insights. The main aim is to check and handle missing values, duplicates, and errors in the data. Python libraries like Pandas, Matplotlib, and Seaborn are used to analyze and visualize the data using graphs. This helps in understanding patterns, trends, and making the data easier to interpret.

Module 1: Data Collection

Dataset Source

The dataset used in this project is the Superstore Sales Dataset, which is a public dataset commonly used for data analysis.

Dataset Description

The dataset contains around 9994 rows and 21 columns, which include details about sales, customers, products, and shipping information.

Key Features

- Order ID, Order Date, Ship Date
- Customer Name, Segment, Region
- Category, Sub-Category, Product Name
- Sales, Quantity, Discount, Profit

Purpose

This dataset is used to analyze sales performance, customer behavior, and business trends using data cleaning and visualization techniques.

Module 2: Data Cleaning & Preprocessing

In this module, the dataset was checked to make sure it is clean and ready for analysis. First, missing values were checked using Python, and no missing data was found. Then duplicate rows were checked, and there were no duplicates present in the dataset.

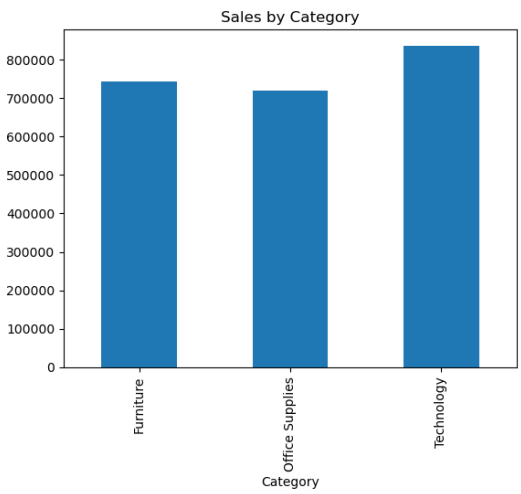
After that, the date columns such as Order Date and Ship Date were converted into datetime format for better analysis. The data types of all columns were also verified. A boxplot was used to check for outliers in the sales column.

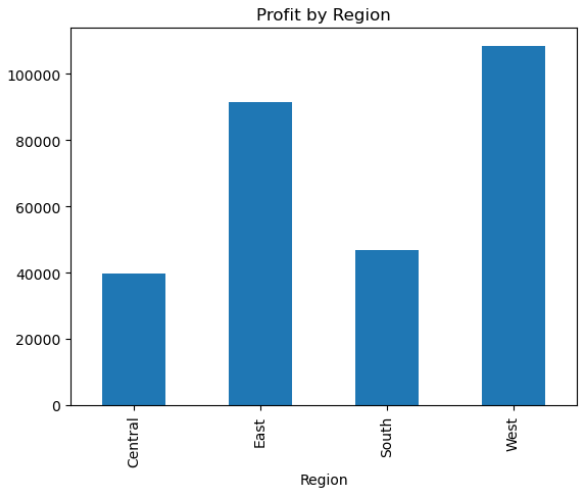
Finally, the cleaned dataset was saved for further use in analysis and visualization.

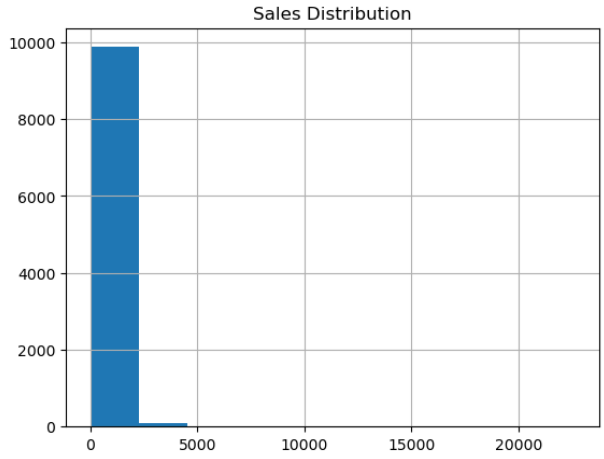
Module 4: Data Visualization

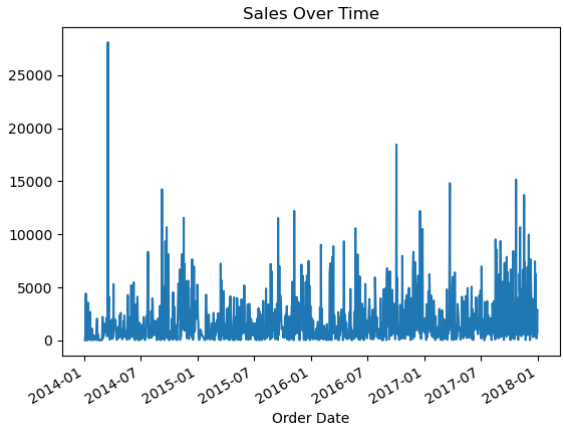
In this module, different graphs were created to understand the data visually. A bar chart was used to compare sales across categories and profit across regions. A histogram was used to show the distribution of sales values. A heatmap was created to understand the relationship between different numerical variables. A line chart was used to analyze sales over time.

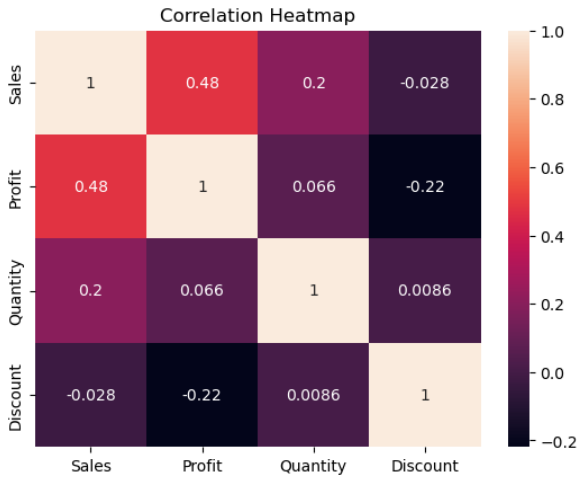
These visualizations help in identifying patterns, trends, and relationships in the dataset in an easy and clear way.

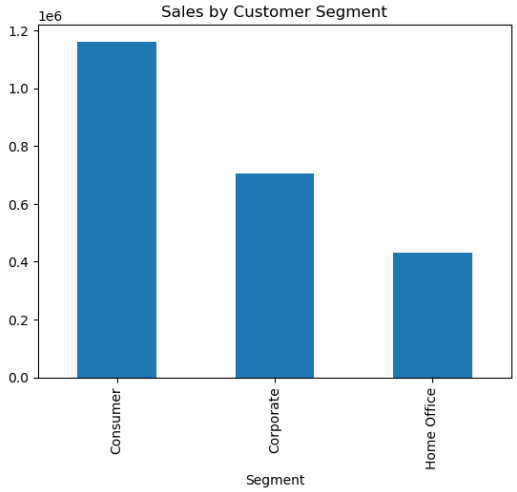












Visual Report of Key Findings

The visualizations created in this project help in understanding the overall performance of the business.

- The bar chart shows that the Technology category has the highest sales.
- The region-wise profit chart indicates that the Western region performs better than others.
- The histogram shows that most sales are in the lower range, with some high-value outliers.
- The heatmap shows the relationship between variables, where discount negatively affects profit.
- The line chart shows variation in sales over time.

These visualizations make it easier to understand patterns and trends in the dataset.

Data Storytelling

From the analysis, it can be understood that the business performs well in the Technology category and in the Western region. However, higher discounts reduce profit, which may affect overall performance. The variation in sales over time shows that business performance is not constant. By focusing on profitable categories and controlling discounts, the business can improve its performance.

Module 5: Dashboard

In this module, a simple dashboard was created using Streamlit to visualize important data. The dashboard includes graphs such as sales by category, profit by region, sales distribution, and correlation heatmap. These graphs help in understanding the data in a clear and interactive way.

The dashboard provides a better view of key findings and makes it easier to analyze business performance. It also helps in quickly identifying trends and relationships in the dataset.

The dashboard was enhanced with a structured layout and multiple visualizations for better understanding of the data.

🚀 **Live Demo**

You can view the deployed dashboard here:
👉 https://data-cleaning-and-visualization-using-superstore-dataset-fmxtz.streamlit.app/

Insights

From the analysis and graphs, some important points were observed. The Technology category has higher sales compared to other categories. The Western region shows better profit performance. It is also noticed that higher discounts can reduce profit in some cases. The sales values are not evenly distributed, and a few high values are present. Sales also vary over time, showing changes in business performance.

Conclusion

In this project, the dataset was cleaned and analyzed using Python. Different graphs were created to understand the data easily. The project helped in finding useful patterns in sales and profit. Overall, this project shows how data cleaning and visualization can help in understanding real-world data.